

Hardware-software integration testing is also essential in connected systems, as issues can emerge at the firmware and connectivity levels even when the hardware is stable.

Repeated testing cycles significantly increase costs. Any design change, component substitution, or hardware fix requires re-testing to ensure system stability, resulting in greater use of lab resources, longer timelines, and increased engineering workload.

Post-failure recovery also incurs costs, as failed units must be analysed, debugged, and requalified before production can resume.

To reduce these risks, engineering teams must create detailed test cases that cover all expected operating conditions. Modern tools, including AI-based testing systems, can help generate test cases from product requirements and improve test coverage. Approaches such as black-box, white-box, functional, and unit testing are used to evaluate products from multiple perspectives. A product should move to production only after successfully completing validation and receiving formal approval from the testing team.

Testing also extends into real-world environments. Products such as GPS-enabled devices often require field testing to verify performance under actual operating conditions. Early deployments of 10–20 units help expose issues not seen in controlled environments, including usability gaps and performance variations.

Pilot production adds another layer of validation. Larger quantities are deployed to end users, and feedback from this stage can lead to design refinements, feature adjustments, and performance improvements. Some changes may require revalidation, including circuit redesigns, PCB layout updates, repeated certifications, and full test cycles, further increasing validation effort and cost.

Manufacturing and scaling costs

The manufacturing phase introduces cost challenges that affect production efficiency, supply-chain stability, and unit economics. A key issue is component availability. Modern electronic products may contain hundreds of components sourced from multiple vendors, and the shortage of even a single critical part can delay production schedules, affecting timelines, product launches, and market entry.

To reduce this risk, engineering teams must plan for component availability during the design phase. Critical components should have approved alternatives wherever possible. If no equivalent component exists, teams must ensure access to reliable distributors and manageable lead times to avoid supply-chain disruption.

However, component replacement is not always straightforward. Even when a substitute matches datasheet specifications, subtle differences can cause system-level failures, leading to rework, scrap risk, and additional engineering effort.

Another cost factor is design for manufacturing (DFM). Products with multiple PCBs, extra fasteners, or complex assembly steps increase assembly time and labour costs. At scale, even small inefficiencies per unit become significant cost drivers. DFM focuses on reducing part count, simplifying assembly steps, and improving production efficiency to lower overall manufacturing costs.

Beyond production efficiency, manufacturing also introduces several operational and post-production cost drivers. Warranty and support responsibilities remain with product companies because contract manufacturers typically do not handle design-related failures. This creates ongoing costs associated with repairs, replacements,

and customer support during the warranty period.

Manufacturing partners also vary in the level of support they provide based on production volume. At higher volumes, they may manage procurement and logistics. At lower volumes, however, product companies often handle supply-chain activities themselves while manufacturers focus solely on assembly.

Each new product introduction also requires training manufacturing teams before production begins. When product lifecycles are short and updates are frequent, these repeated training cycles reduce the time available to recover setup costs.

Finally, scaling introduces additional validation risks. Systems that work at small deployment volumes may fail at large scale. Infrastructure components, such as message brokers and cloud services, must be validated under higher loads early in the process to avoid costly redesigns later.

Post-deployment operational lifecycle cost

Connected devices must be designed to operate in environments with unstable network conditions. Intermittent connectivity is common, so systems need clear strategies for handling data during outages. This includes deciding what information should be stored locally and what can be safely discarded. Local buffering ensures that critical data is not lost, while non-essential or redundant data can be filtered out to avoid unnecessary storage use. In parallel, firmware update systems must account for devices that miss updates due to connectivity gaps, ensuring mechanisms to detect, retry, and track update status across distributed fleets.

As connected products continuously generate data, cloud and storage costs can grow quickly. Managing these costs depends on selecting data at the source. Instead